

FairCredit Jurisdiction Navigator - Plain-Language Summary

FairCredit Jurisdiction Navigator is a prototype jurisdiction-aware RegTech tool designed for American Express’s AI-supported credit card approval process. It does not replace the credit model or make final lending decisions. Instead, it works as a compliance and governance layer above the model.

The main idea is: The same AI credit decision should not have the same compliance meaning in every jurisdiction.

The prototype compares two jurisdictions: the United States and the European Union.

I used 2,000 synthetic credit application records. No real American Express customer data was used. The dataset includes variables such as income, debt-to-income ratio, credit history length, missed payments, credit utilisation, employment status, spending volatility, and requested credit limit. A synthetic_group_proxy variable is included only for fairness monitoring and is not used to train the model.

The prototype uses a simple logistic regression model to estimate whether an applicant is likely to show good repayment behaviour. The model produces three possible outcomes:

Output	Meaning
Approved	Applicant is above the approval threshold
Manual Review	Applicant is close to the decision boundary
Declined	Applicant is below the approval threshold

In the test sample of 600 applications, the model approved 500, sent 50 to manual review, and declined 50.

The tool then applies different jurisdictional logic.

Mode	Main Focus	Key Output
US Mode	Individual adverse action explanation	Specific reasons for decline or manual review
EU Mode	High-risk AI governance	Fairness monitoring, drift detection, human review trigger

In US mode, the tool generates plain-language adverse action reasons, such as high debt-to-income ratio, limited credit history, recent missed payments, or high credit utilisation. The key question is: Can the firm explain this specific negative credit decision clearly and accurately?

In EU mode, the tool monitors system-level risks. In the prototype, Group B had the lowest approval rate at 73.96%, compared with 88.15% for Group A. The approval-rate disparity ratio was 0.839, above the prototype alert threshold of 0.80. Therefore, no hard fairness alert was triggered, but Group B should remain monitored.

The stronger governance signal came from drift monitoring. The Population Stability Index for debt-to-income was 0.237, above the review threshold of 0.20. This means the EU mode would trigger human model governance review.

The tool has clear limits. It uses synthetic data, simplified thresholds, and a simple model. It does not provide legal advice, reproduce American Express’s real credit policy, or make final approval decisions. Its value is to make jurisdiction-specific compliance questions visible and route uncertainty to human review.